Business Consultancy Report

Data Science 3000B – Introduction to Machine Learning

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Jacob Hills

Anhad Kaur

John-Anthony Mastrogiacomo

Seongmin Park

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Executive Summary

Canadian household expenditure data reveals that housing burden, measured as the proportion of disposable income allocated to shelter, is not primarily a function of geographic or demographic segmentation but of how households distribute spending across necessity and discretionary categories. This finding has direct implications for how affordability policy is designed and who it targets.

Using K-Means clustering and Principal Component Analysis, dissemination areas were grouped into two segments: a small cluster of high-volume urban areas representing dense, high-spending neighbourhoods, and a large cluster of typical households representing the broad Canadian norm. Critically, median housing burden is nearly identical across both groups, confirming that expenditure scale alone does not predict financial strain.

Predictive modelling using both an Elastic-Net generalized linear model and an XGBoost regressor found that spending composition, particularly the near-absence of discretionary and recreational expenditure, is the strongest signal of housing burden. XGBoost substantially outperformed the linear model for urban areas, achieving an R^2 of 0.933 on that segment compared to near zero for the linear approach, confirming that housing burden dynamics in dense areas are governed by non-linear interactions. SHAP analysis identified recreational vehicle spending, necessity share, and food share as the primary drivers of predicted burden.

The central recommendation of this report is that housing affordability assessments should incorporate spending composition data alongside traditional shelter cost metrics, and that XGBoost should be adopted as the primary predictive tool for identifying at-risk dissemination areas, particularly in urban contexts.

Background

Project Scope

This project examines how Canadian households allocate their disposable income toward shelter-related expenditures at the dissemination area level. The primary objective is to identify patterns, disparities, and key factors of housing cost burden across geographic and demographic segments.

Definitions / Data Understanding

A key metric used throughout this analysis is the Housing Burden Ratio (HBR), which is defined as following:

$$HBR = \frac{HSSH001S}{HSAGDISPIN}$$

where $HSSH001S$ represents household expenditure summary for shelter-related category, and $HSAGDISPIN$ represents the household disposable income. Also note that term HBR and y may be interchangeably used throughout this report.

Our tableau has various consumption categories such as food; household operations; transportation; health care; recreation; clothing; education; and more. See part 1 (c) in Jupyter Notebook (.ipynb) file included for the full list of these categories.

Also, the tableau includes series of "_share" variables as:

$$\text{share}_i = \frac{\sum_i \text{expenditure}_i}{\text{total consumption}}$$

where i is the consumption categories. These variables represent the proportion of total household consumption allocated to specific expenditure categories. These variables capture relative spending behaviour, which allows us to draw meaningful comparisons across dissemination areas with different income levels.

For our analysis, these consumption shares are interpreted as indicator of household financial priorities and constraints. A key hypothesis is that households allocating a higher proportion of spending toward essential goods are more likely to experience tighter budget constraints and consequently, higher HBR.

Then, each category is groups into one of two groups. Necessities, where all essential categories for basic needs are grouped. And discretionary, where categories where the disposable income can be spent more flexibly than necessities. Under this framework, higher necessity shares may signal lower disposable income, leaving less flexibility after housing costs. And higher discretionary shares may signal greater financial flexibility, potentially correlated with lower housing burden.

Research Methods

A combination of exploratory data analysis, dimensionality reduction, clustering, and predictive modeling techniques were used to carry out this report:

- K-Means clustering to identify distinct groups of dissemination areas with similar spending and demographic characteristics,
- Principal Component Analysis (PCA) to reduce dimensionality and uncover key factors driving variability,
- Uniform Manifold Approximation and Projection (UMAP) to visualize complex, non-linear structure in the data,
- Elastic-Net Generalized Linear Model (Elastic-Net GLM) to model relationships between predictors and shelter expenditure while managing multicollinearity,
- XGBoost to capture non-linear interactions and improve predictive performance, and
- Shapley Additive Explanations (SHAP) to interpret model outputs and quantify feature performance.

Findings

Data Distribution

After splitting the dataset into test and training sets, we examine the distribution of the response variable, HBR.

Statistic	Value
Count	45485
Mean	0.257026
Standard Deviation	0.058526
Minimum	0.11686
25th Percentile	0.222215
50th Percentile (Median)	0.251285
75th Percentile	0.283068
Maximum	4.494179

Table 1. Distribution of Training Set of HBR

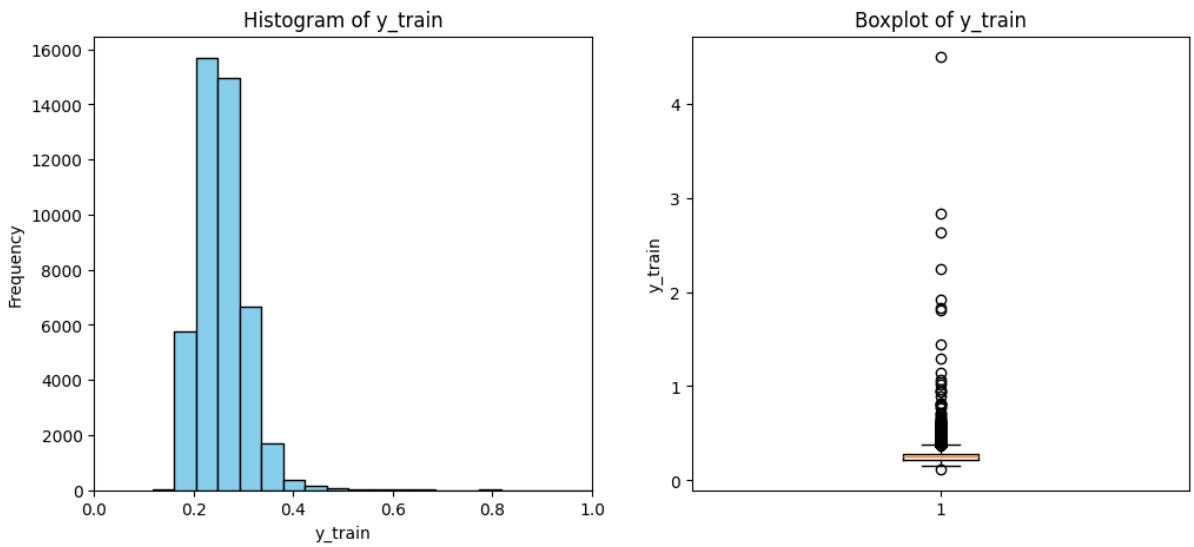


Figure 1. Histogram and Boxplot of Training Set of HBR

From above table and figures, we can see that the distribution of training set of HBR is skewed and is not evenly distributed.

Then, before we proceed with any data analysis, we calculate the correlations between each variable and HBR to get the top 10 predictor for HBR.

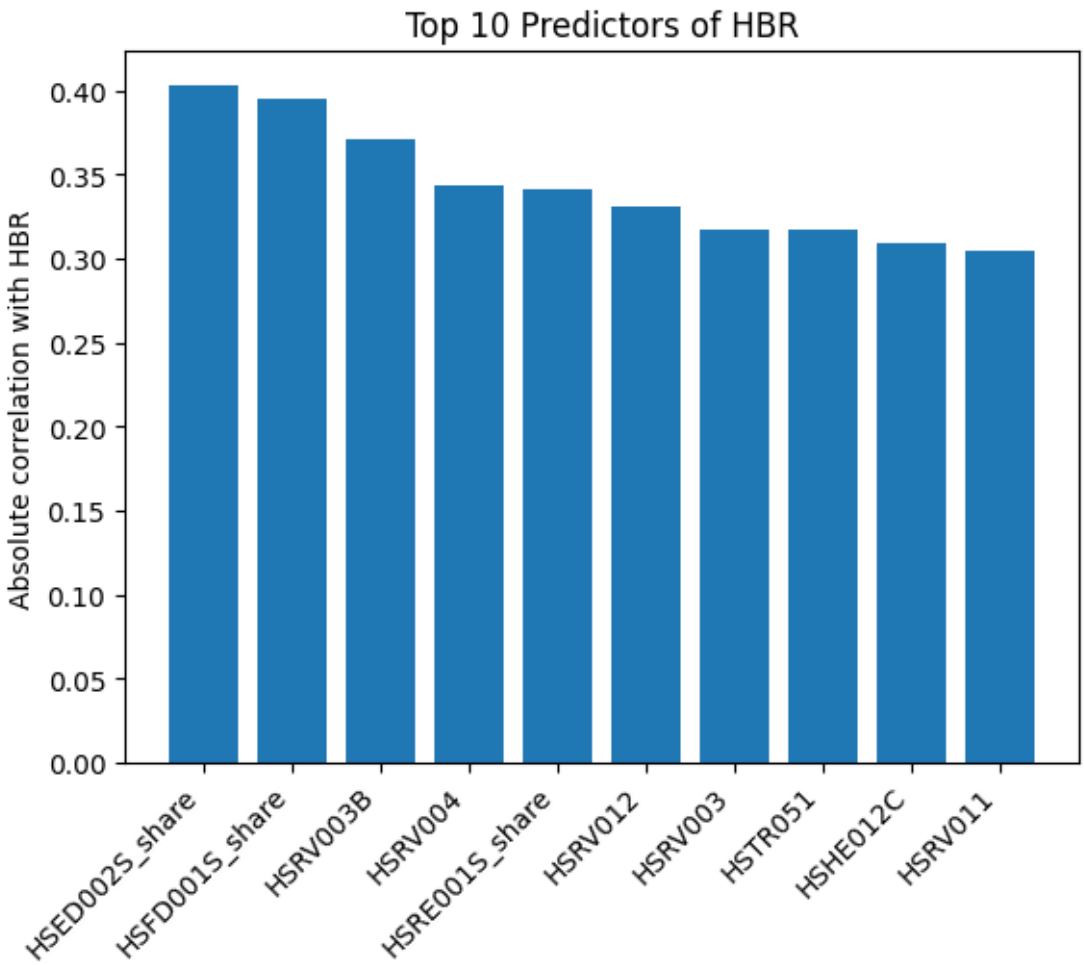


Figure 2. Top 10 Predictor of HBR based on Absolute Correlation

K-Means Clustering

K-Means clustering was applied to the standardized training set for values of k ranging from 2 to 12. The silhouette score peaked at 0.786 for k = 2 and declined monotonically for all higher values, while the elbow curve showed its steepest inertia drop between $k = 2$ and $k = 3$, with diminishing returns thereafter. Both methods agree on $k = 2$ as the optimal number of clusters.

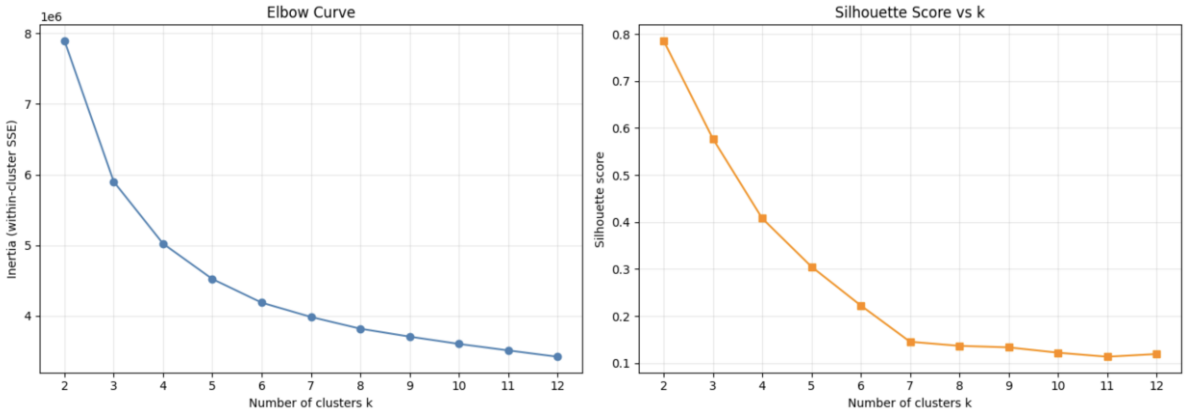


Figure 3. Elbow Curve and Silhouette Score of K-Means Clusters

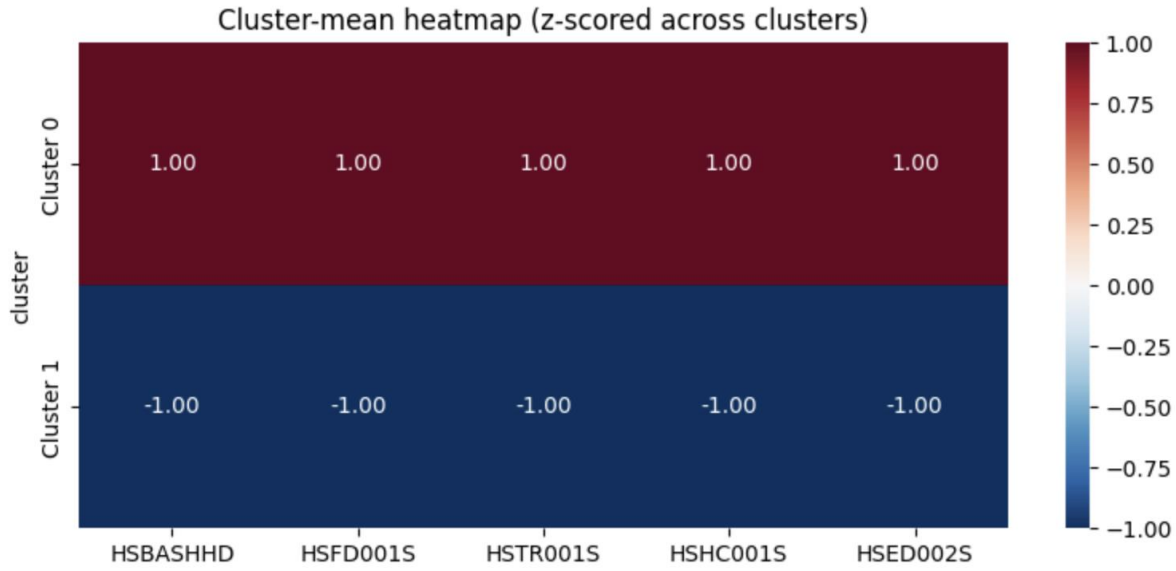
It is worth noting that z-score standardization and median imputation tend to compress the feature space uniformly, which can suppress finer cluster structure and bias the optimum toward smaller values of k. Alternative preprocessing such as log-transformation or PCA pre-processing prior to clustering would likely shift the optimal k upward by introducing more differentiated local structure.

The two clusters differ markedly in scale. Cluster 0 contains 1,166 dissemination areas with a mean household count of approximately 1,265, compared to 260 for Cluster 1. This elevated profile extends consistently across food, transportation, health care, and education spending. Cluster 0 is designated High-Volume Urban Areas, reflecting densely populated dissemination areas with elevated absolute expenditures. Cluster 1 contains 44,319 observations representing the vast majority of Canadian dissemination areas and is designated Typical Households.

```
Cluster sizes:
cluster
0     1166
1    44319

Cluster means (raw scale):
         HSBASHHD     HSFD001S     HSTR001S    HSHC001S    HSED002S
cluster
0         1265.14  17658443.95  18017112.33  4368738.13  3516968.53
1          260.04   3214752.80   3219187.36   827512.65   425331.56
```

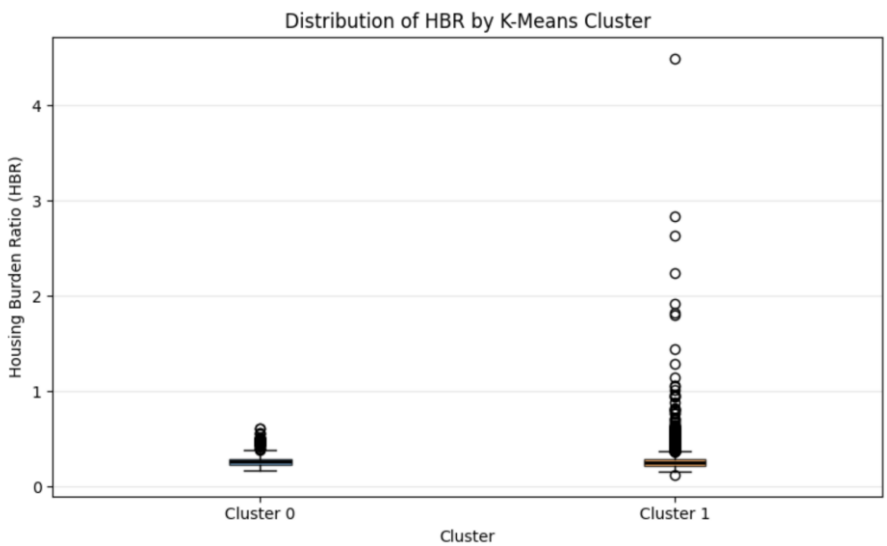
Table 2. Cluster Sizes and Cluster Means




```
Cluster names:
  0: High-Volume Urban Areas
  1: Typical Households
```

Figure 4. Cluster-mean Heatmap

Examining the distribution of HBR by cluster reveals nearly identical medians of 0.259 for Cluster 0 and 0.251 for Cluster 1, with comparable interquartile ranges of approximately 0.060 in both cases. The clusters therefore do not capture meaningful differences in housing burden itself. Housing cost pressure appears broadly uniform across both segments, indicating that the clustering structure is driven by expenditure volume rather than financial strain.




```
Cluster 0: median=0.2590  IQR=0.0603  n=1166
Cluster 1: median=0.2511  IQR=0.0609  n=44319
```

Figure 5. Boxplots of Each Cluster

PCA

Principal Component Analysis was performed on the standardized training set to uncover the latent linear structure of household expenditure patterns. The first principal component alone explains 72.6% of total variance, with the top ten components cumulatively accounting for 93.1%. The full breakdown is as follows: PC1 at 72.6%, PC2 at 78.3%, PC3 at 81.9%, PC4 at 84.5%, PC5 at 86.7%, PC6 at 88.6%, PC7 at 90.1%, PC8 at 91.2%, PC9 at 92.2%, and PC10 at 93.1%. The sharp dominance of the first component indicates that a single axis, broadly interpretable as overall expenditure scale, accounts for most of the variation across Canadian dissemination areas.

```
Cumulative variance explained by top 10 PCs:
  PC1: 72.56%
  PC2: 78.27%
  PC3: 81.91%
  PC4: 84.48%
  PC5: 86.70%
  PC6: 88.55%
  PC7: 90.06%
  PC8: 91.23%
  PC9: 92.21%
  PC10: 93.08%
```

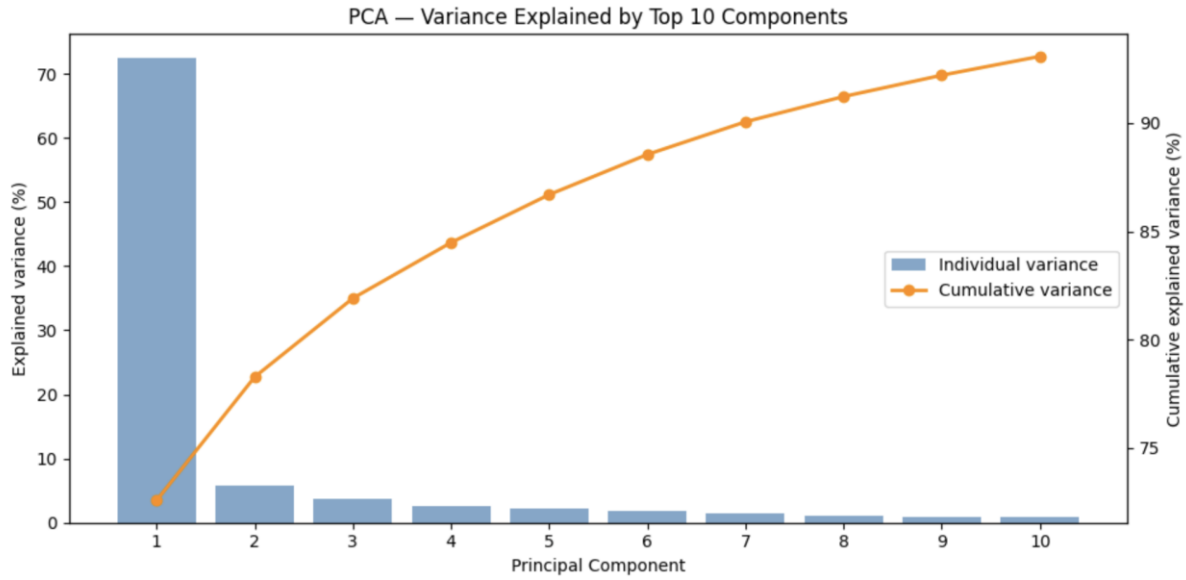


Figure 6. Top 10 Components' Variances by PCA

A scatter plot of PC1 versus PC2 coloured by K-Means labels reveals that the two clusters are cleanly separated along the PC1 axis. Cluster 0 occupies distinctly high values of PC1 while Cluster 1 concentrates tightly near the origin. This linear separability confirms that the data satisfies the cluster assumption and that the structure identified by K-Means reflects a genuine and dominant gradient in the data rather than an algorithmic artifact.

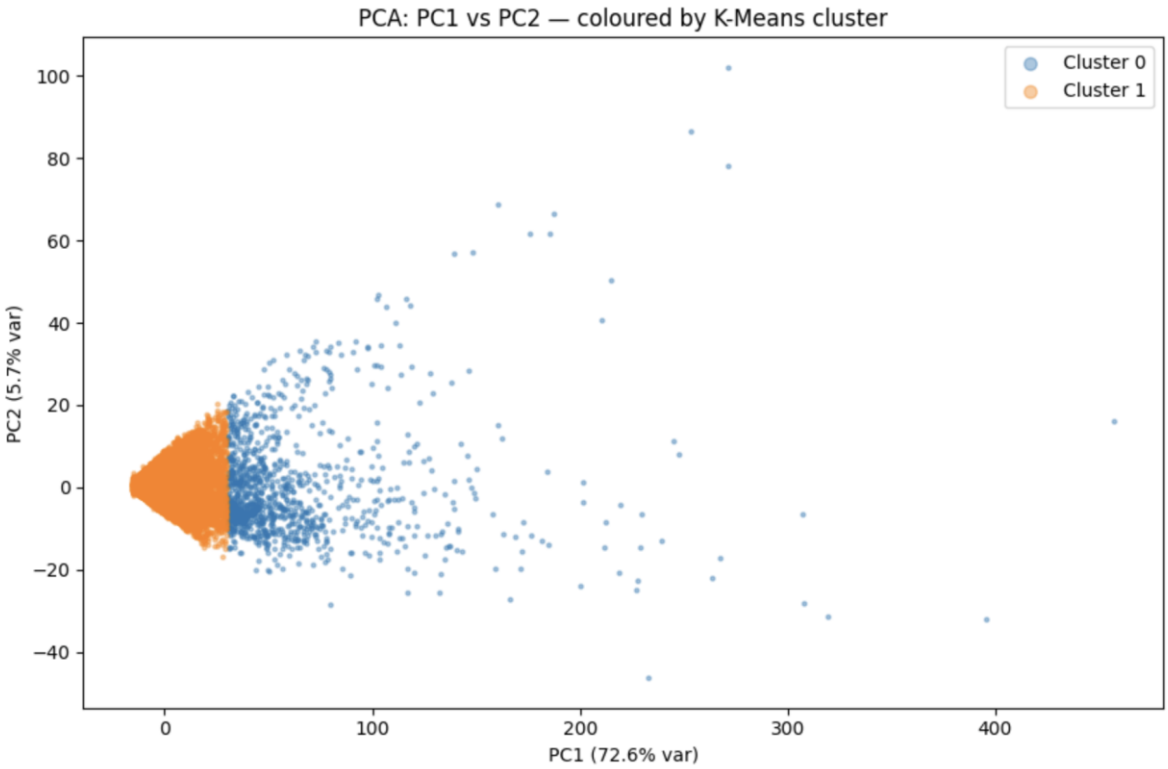


Figure 7. PC1 vs. PC2 Coloured by K-Means Clusters

Examining the top-loading features for each of the first three components provides further interpretive depth. PC1 is dominated by total consumption, household operations (HSHO001S), and recreation (HSRE001), all loading positively at around 0.07, and represents overall expenditure scale. PC2 contrasts transportation (HSTR051, positive at 0.178) against health and recreational vehicle expenditures (HSHE012C at negative 0.158, HSRV011 at negative 0.157), capturing a compositional trade-off between transportation-heavy and recreation-heavy spending profiles. PC3 is characterized by positive loadings on recreational spending share (HSRV005M1 at 0.160, HSRE001S_share at 0.157) and negative loadings on transportation sub-categories (HSTR013 at negative 0.141), capturing a further axis of discretionary spending composition.

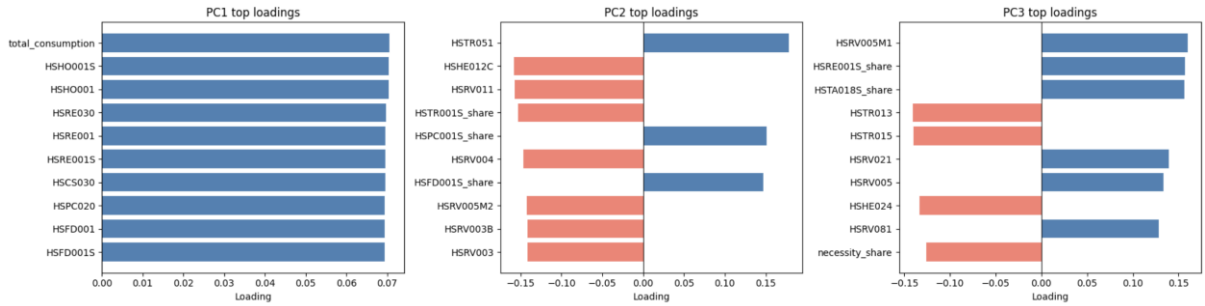
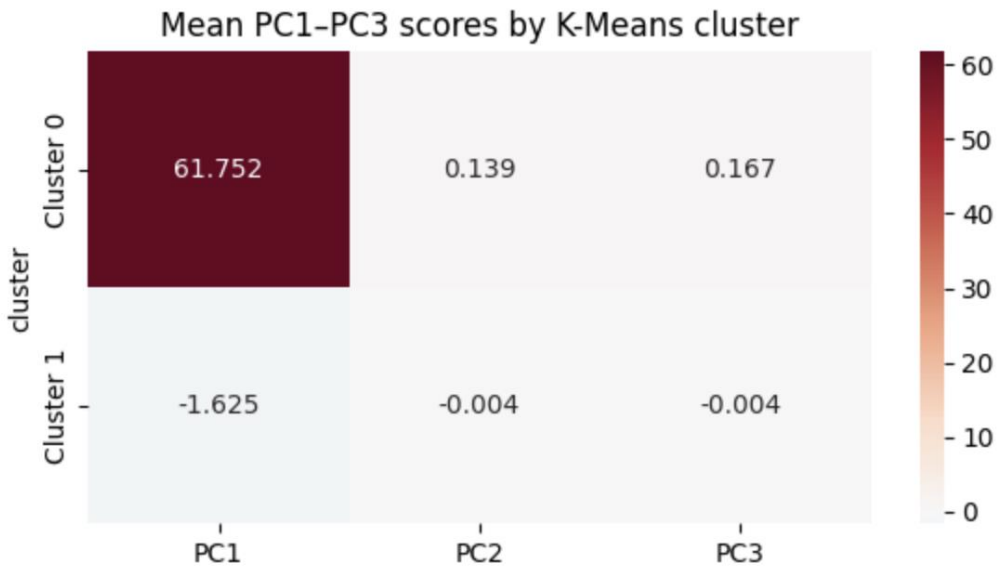


Figure 8. Top Loading Features for PC1, PC2, and PC3

Mean PC scores by cluster confirm the above interpretation. Cluster 0 has a mean PC1 score of 61.75 compared to negative 1.63 for Cluster 1, while PC2 and PC3 scores are near zero for both groups. Based on this analysis, Cluster 0 is re-designated High-Scale Expenditure and Cluster 1 is re-designated Baseline Expenditure. These names differ from the K-Means designations by grounding the clusters in the latent structure of the data rather

than raw variable comparisons, making them more analytically precise. The silhouette score in the PCA two-dimensional space is 0.831, exceeding the original high-dimensional score of 0.786 and confirming that PCA not only preserves but enhances cluster separability.

```
Mean PC scores by cluster:
            PC1    PC2    PC3
cluster
0        61.752  0.139  0.167
1        -1.625 -0.004 -0.004
```




```
Revised cluster names from PCA analysis:
  Cluster 0: High-Scale Expenditure (high PC1, dense urban areas)
  Cluster 1: Baseline Expenditure (near-zero on all PCs, typical households)
```

Figure 9. Mean PC Scores by Cluster

UMAP

The UMAP dimensionality reduction analysis performed revealed many important limitations in its representation of the structure of Canadian household data. Although UMAP is designed to preserve local relationships in high-dimensional data, the 2-dimensional embedding performed clearly did not separate the clusters identified through K-Means, as seen in the K-Means coloured scatterplot. This was further reflected in the silhouette score, dropping from 0.786 in the original feature space to just 0.234 in the UMAP space, indicating a significant loss of cluster separability.

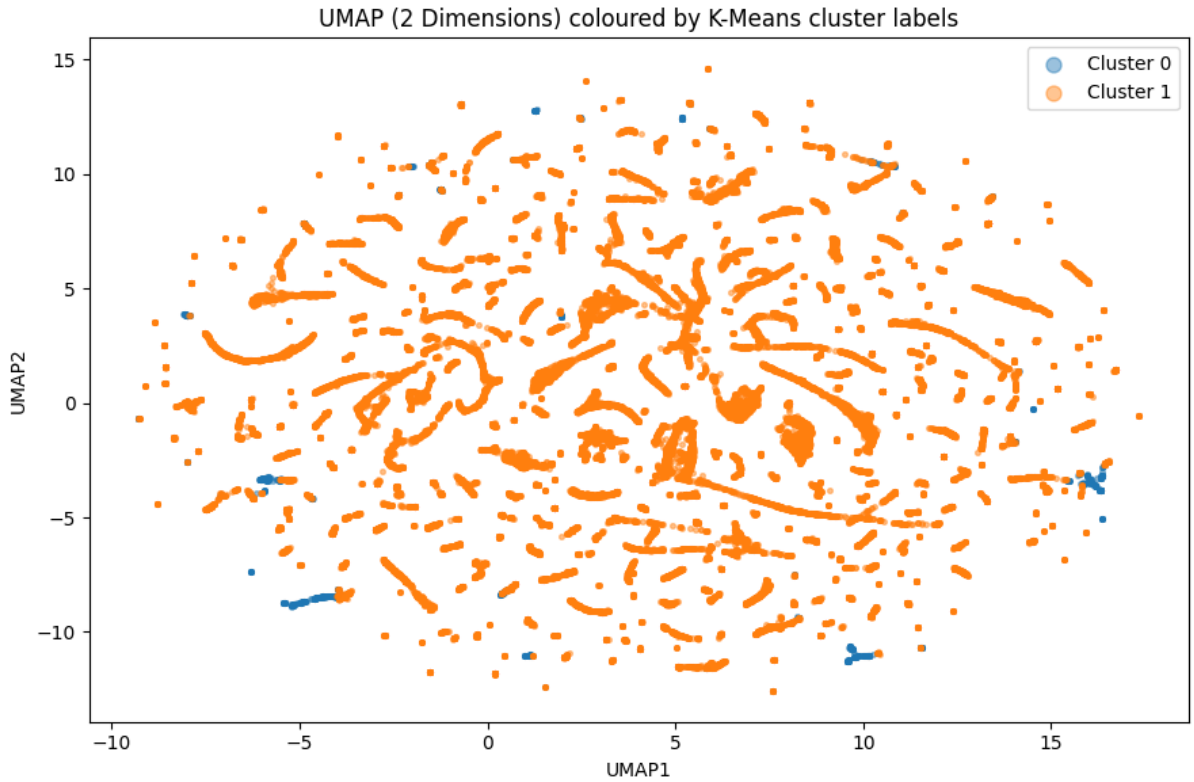


Figure 10. UMAP Projection on K-Means Clusters

Looking more closely at the UMAP projection on K-Means clusters, there is one dominant cluster occupying most of the space, with the second appearing only in sparse regions. These findings suggest the structure identified by K-Means is primarily driven by global variance rather than localized structure. The PCA cluster separation preserved cluster separation much more effectively, achieving a silhouette score of 0.831. This indicates that the dataset is heavily influenced by linear relationships rather than complex non-linear correlations.

These findings overall imply that household expenditure patterns are influenced by broad, continuous gradients rather than sharply defined local groups, Limiting the usefulness of non-linear embedding techniques for segmentation. Therefore, we may want to look at simple linear representations as being more appropriate for interpretation and decision making.

Elastic-Net GLM

The Elastic-Net Generalized Linear Model helped to provide further insight towards the key driving factors of housing burden. By transforming the target variable using a Pearson Type III distribution and mapping it to a normal space, the model was able to handle skewness in the Housing Burden Ratio effectively. As seen from the summary of the transformed data, we produced a distribution very close to standard normal, enabling us to assume stable model convergence.

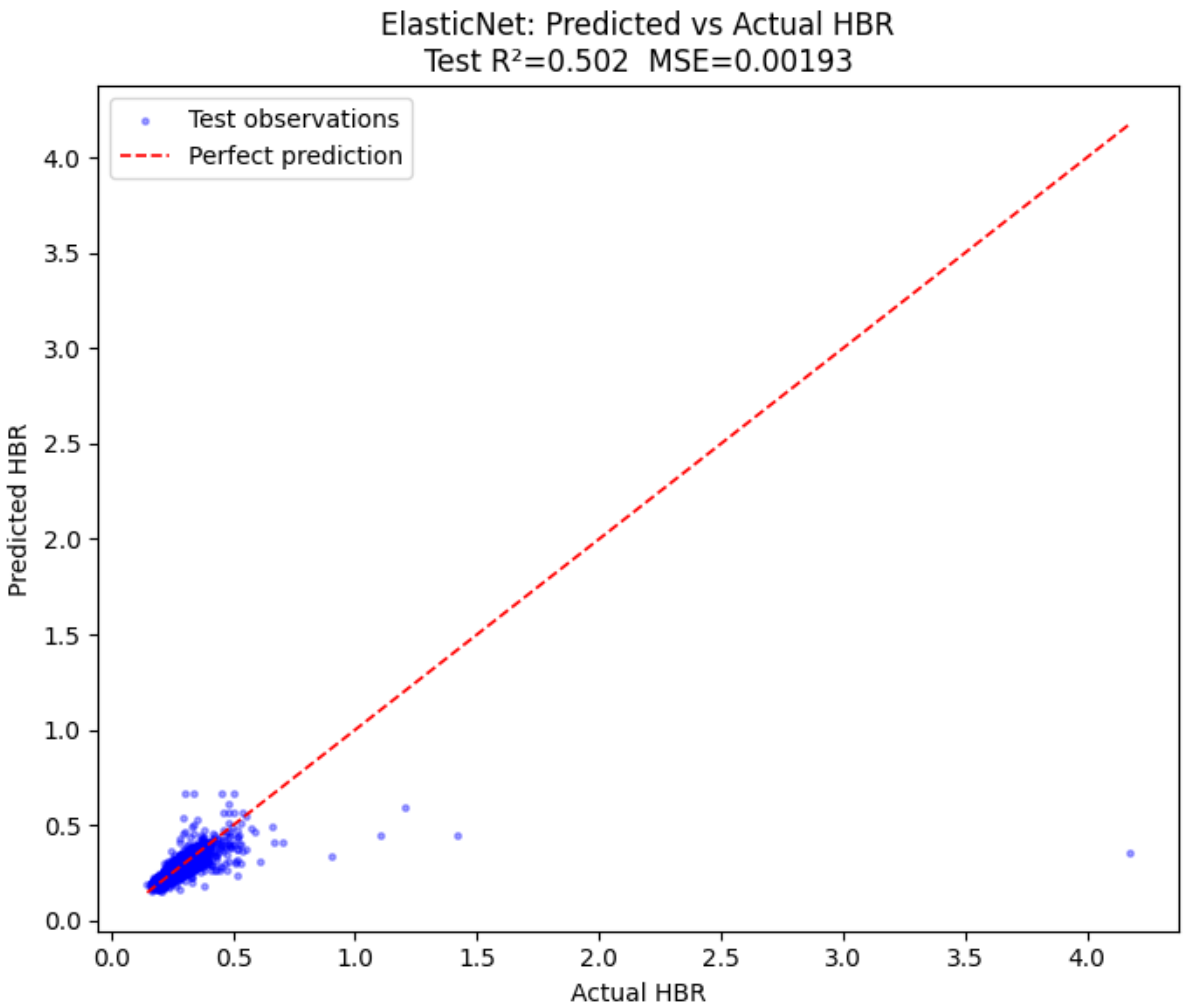


Figure 11. Predicted vs. Actual HBR (ElasticNet)

The Elastic-Net GLM model achieved a test R^2 of approximately 0.502 (95% CI [0.278, 0.778]). These values indicate moderate predictive performance, suggesting that while the model captures the key drivers of housing burden, there remains a substantial unexplained amount of variability. The low test mean squared error (MSE) value reflects accurate predictions for typical observations but weaker performance on extreme values.

Cluster	R^2	95% Confident Interval
0	0.027378	[-0.603630, 0.404572]
1	0.512386	[0.311114, 0.804682]

Table 3. Per-cluster R² with 95% CI

Looking at the cluster-level analysis, there are significant differences in model performance between clusters. The Elastic-Net model performs substantially better for the dominant cluster ($R^2 \approx 0.512$) than for the smaller cluster ($R^2 \approx 0.027$), suggesting that the relationships between the predictors and housing burden differ across the two segments. This is consistent with the earlier findings where one cluster represented the vast majority of households and the other captured atypical or high-variance cases

An important note is that the Elastic-Net model retained 234 non-zero coefficients out of the 274, suggesting that housing burden is influenced by a wide variety of factors rather than a small subset of dominant predictors. The optimal model (l1_ratio = 1.0) indicates that a LASSO-like solution was preferred, suggesting that feature selection is in fact critical despite the large number of predictors. The most important coefficients were associated with food expenditure, household size, and other consumption-related variables.

As a whole, the Elastic-Net GLM demonstrates that housing burden largely follows broad linear trends, however its drivers vary significantly across household segments.

XGBoost

XGBoost was trained directly on the original HBR values without distributional transformation, as tree-based models make no assumptions about the response distribution. The parameter grid spanned n_estimators from 100 to 500, max_depth from 3 to 9, learning rate from 0.01 to 0.2, subsample from 0.7 to 1.0, and colsample_bytree from 0.7 to 1.0. These ranges were chosen to balance model complexity against overfitting risk across approximately 45,000 training observations and 274 features. RandomizedSearchCV with 3-fold cross-validation and 5 iterations was used for computational feasibility. The optimal configuration was n_estimators = 300, max_depth = 7, learning_rate = 0.05, subsample = 0.9, and colsample_bytree = 0.7, yielding a cross-validated R^2 of 0.677.

On the held-out test set, XGBoost achieved an R^2 of 0.544 and an MSE of 0.0018, with a bootstrapped 95% confidence interval of [0.376, 0.832]. Compared to the Elastic-Net ($R^2 = 0.502$, 95% CI [0.278, 0.778]), XGBoost delivers a higher point estimate, a higher lower bound, and a narrower interval, indicating both greater accuracy and greater consistency across bootstrap samples.

```
Bootstrapped 95% CI for R²: 0.376, 0.832
```

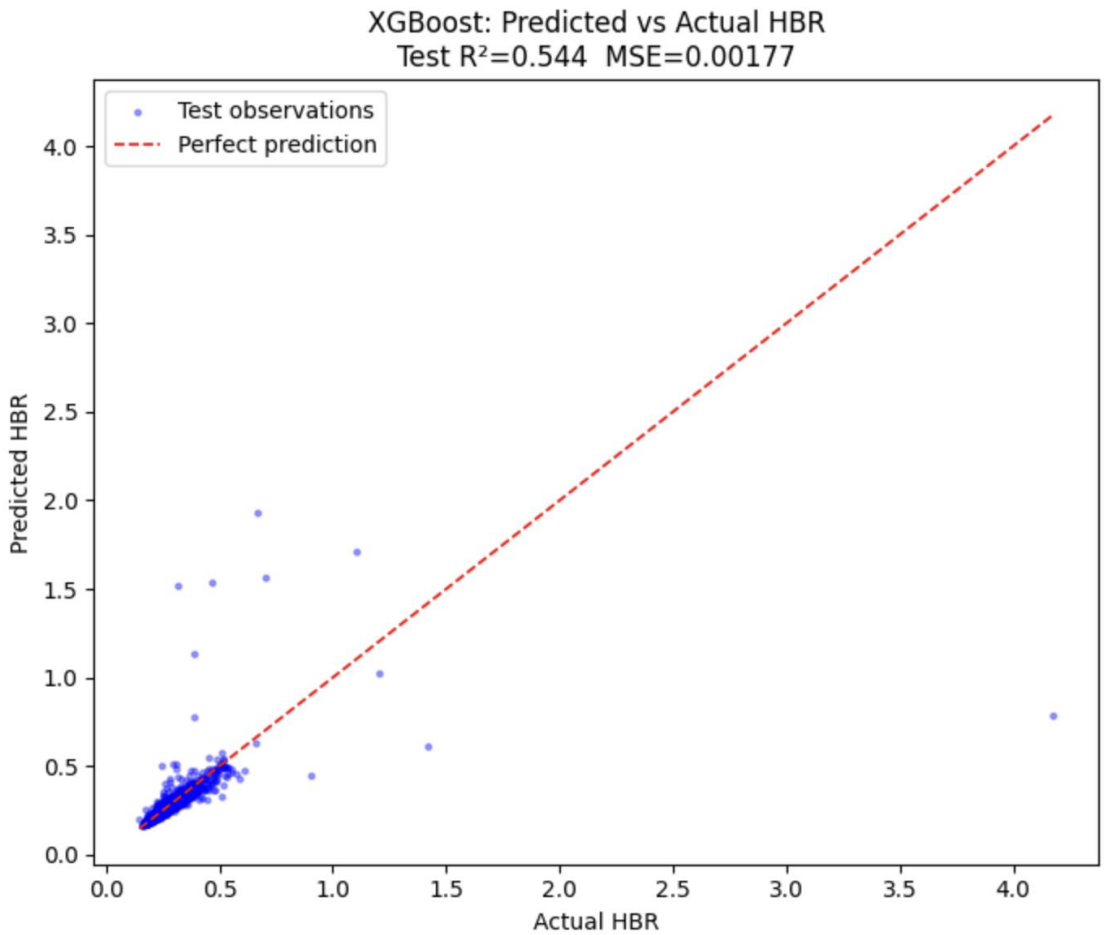


Figure 12. Predicted vs. Actual HBR (XGBoost)

The per-cluster breakdown reveals the most decisive difference between the two models. For Cluster 0 (High-Volume Urban Areas), XGBoost achieves an R^2 of 0.933 compared to 0.027 for the Elastic-Net on the same cluster. For Cluster 1 (Typical Households), performance is comparable at 0.535 versus 0.512. This confirms that the housing burden relationship is broadly linear for typical households but strongly non-linear for high-volume urban areas, where threshold effects and expenditure interactions dominate.

```
Per-cluster R² with 95% CI:
 cluster       r2   CI_low  CI_high
       0 0.933172 0.908348 0.950193
       1 0.534837 0.363640 0.827201
```

Table 4. Per-Cluster R² with 95% CI

SHAP Explainer

SHAP values were computed for the XGBoost model using a TreeExplainer, providing a consistent decomposition of each feature's contribution to individual predictions. In the context of this problem, each SHAP value quantifies how much a given expenditure variable shifts the predicted HBR above or below the baseline expectation for a dissemination area.

The bee-swarm summary plot reveals wide distributions of SHAP values across observations, with clear directional patterns for the most impactful features. The top 10 features by mean absolute SHAP are HSRV003B (0.01570), HSBASHHD (0.00676), HSHE012C (0.00481), HSHO002 (0.00472), HSWH040S (0.00445), HSFD001S share (0.00392), necessity share (0.00329), HSWH028S (0.00282), HSEP011S (0.00255), and HSTR055 (0.00234). The prominence of recreational vehicle spending, necessity share, and food share among the top drivers is economically interpretable. Dissemination areas with little discretionary spending are more likely to be budget-constrained, leaving a higher proportion of income absorbed by shelter costs, while areas with substantial recreational and education expenditure signal greater financial flexibility.

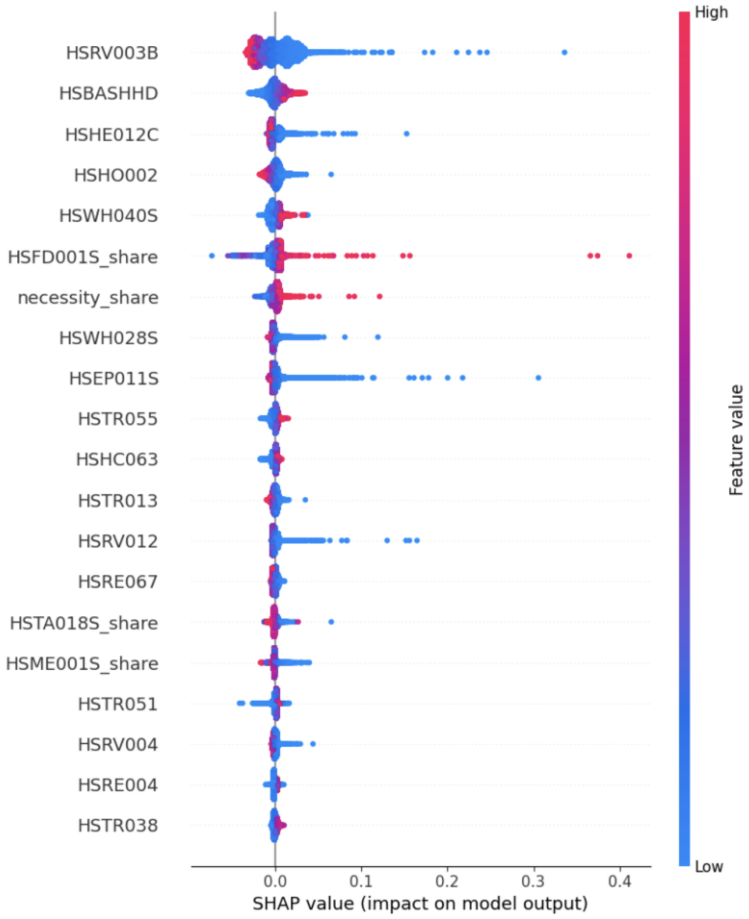


Figure 13. Summary Plot of SHAP Values

Two dependence plots were generated for the top two features. HSRV003B was selected because it is the single most impactful predictor and exhibits a strongly non-linear relationship with predicted burden. Dissemination areas with near-zero recreational vehicle expenditure receive large positive SHAP contributions, indicating higher predicted burden, while the contribution drops sharply toward zero as spending increases, consistent with a threshold effect. This pattern suggests that the absence of recreational spending acts as a signal of household financial constraint rather than a continuous linear driver.

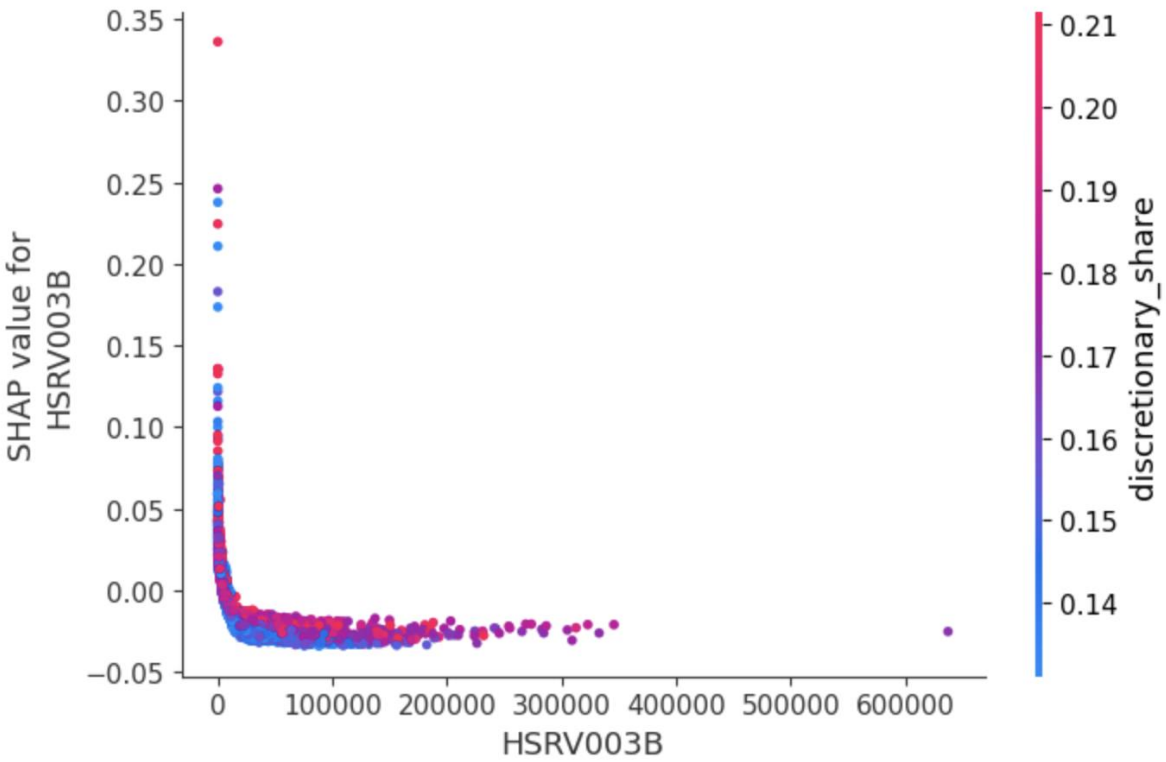


Figure 14. Dependence Plot (HSRV003B)

HSBASHHD was selected because it bridges the clustering and predictive analyses. Its dependence plot shows a positive but quickly saturating relationship, where larger dissemination areas contribute modestly positively to predicted burden, reflecting the urban density effect captured by Cluster 0. Health spending patterns co-vary with area density in a way that further modulates the burden prediction.

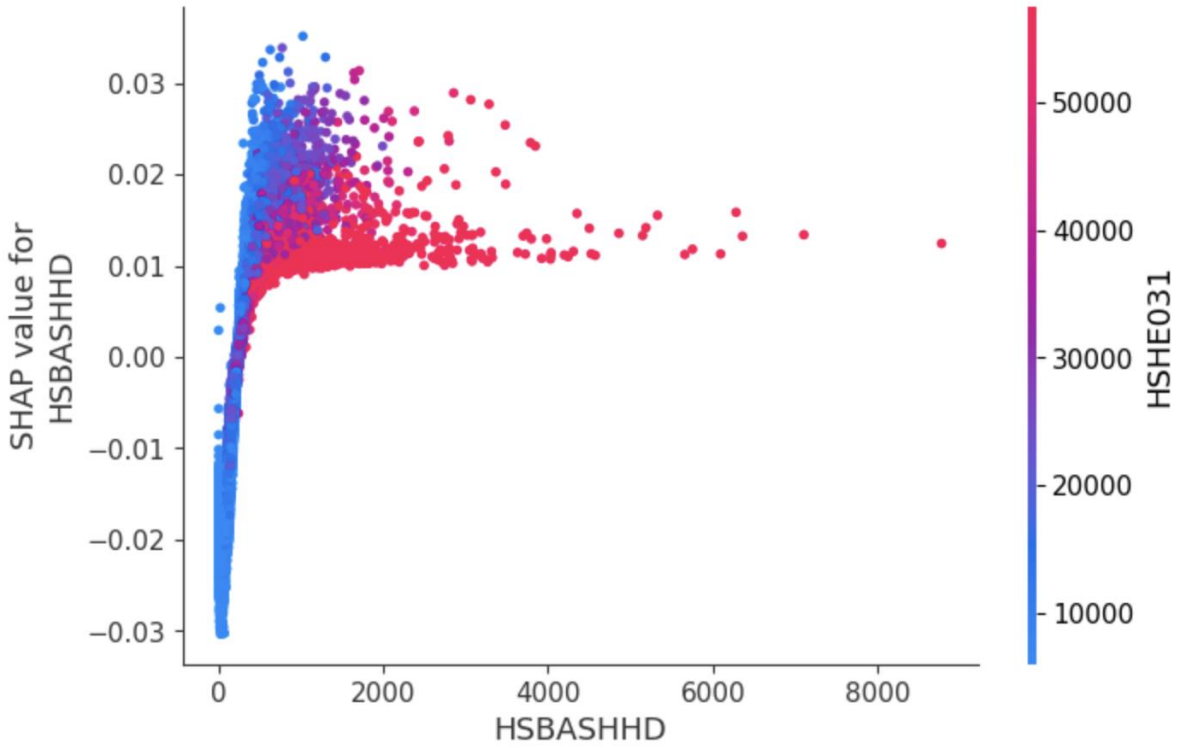


Figure 15. Dependence Plot (HSBASHHD)

Comparing the SHAP-identified drivers to the Elastic-Net coefficients reveals meaningful divergence. The Elastic-Net highlighted food expenditure volume and household operations as the dominant linear predictors, while SHAP elevates recreational vehicle spending and household count, capturing threshold-based interactions a linear model cannot represent. These approaches are complementary: the linear model identifies primary additive drivers while SHAP reveals the non-linear structure explaining residual variance, particularly within the High-Scale Expenditure cluster. Examining HBR distributions across clusters confirms that median burden is nearly identical between groups at 0.259 and 0.251, establishing that the features separating the clusters along PC1 are not the same features driving burden within them. Clustering captures between-area differences in expenditure volume and urban density, while the supervised models capture within-area variation in spending composition. Both perspectives are necessary for a complete understanding of housing affordability in Canada.

Recommendations

K-Means Clustering

To improve the identification of at-risk households, we recommend supplementing or replacing absolute-expenditure-based clustering with a composition-based approach, because the current clusters reflect expenditure scale rather than housing affordability and produce nearly identical median HBR values across groups. Applying clustering after log-transformation of spending variables, or restricting features to expenditure shares rather than totals, is likely to produce segments with more genuinely distinct affordability profiles and greater policy utility.

PCA

To support downstream modelling and segmentation work, we recommend incorporating the first principal component as a summary feature in any further analysis, because it captures nearly three quarters of total variation and cleanly distinguishes urban from typical household areas. Analysts using this data for affordability policy should be aware that the dominant axis of variation reflects spending power and urban density rather than housing stress, and should supplement PCA-derived features with spending composition variables whenever affordability is the primary concern.

UMAP

To avoid drawing incorrect conclusions about household group structure, we recommend against using UMAP projections as the basis for segmentation decisions in this context, because the two-dimensional embedding loses most of the cluster separability present in the original feature space. Full-dimensional analyses or structure-preserving methods such as PCA should be prioritized, and any low-dimensional visualizations should be treated as exploratory tools rather than decision inputs.

Elastic-Net GLM

To identify the broad linear drivers of housing burden in a transparent and communicable way, we recommend retaining the Elastic-Net model as an interpretive complement to XGBoost, because it clearly surfaces the primary additive predictors such as food expenditure and household size. However, a segmented modelling strategy should be adopted rather than applying a single generalized model across all dissemination areas, as the drivers differ significantly between the high-volume urban and typical household segments and a pooled approach obscures important within-group variation.

XGBoost

To maximize predictive accuracy for housing burden estimation, particularly in urban dissemination areas where policy interventions carry the greatest consequence, we recommend adopting XGBoost as the primary production model, because it achieves an R^2 of 0.933 for the high-volume urban cluster compared to near zero for the linear alternative. The model should be retrained periodically as new expenditure cycles become available, and the parameter search grid should be expanded beyond the five candidates evaluated here when computational resources allow.

SHAP Explainer

To build a more proactive and accurate early warning system for housing financial strain, we recommend that affordability assessments incorporate the absence of discretionary spending as a monitoring signal alongside traditional shelter cost metrics, because near-zero recreational vehicle expenditure is the single strongest predictor of elevated housing burden in the XGBoost model. Because the features driving housing burden differ substantially from those separating geographic clusters, any policy framework should address spending composition and expenditure volume as separate dimensions rather than assuming that segment membership captures affordability risk.